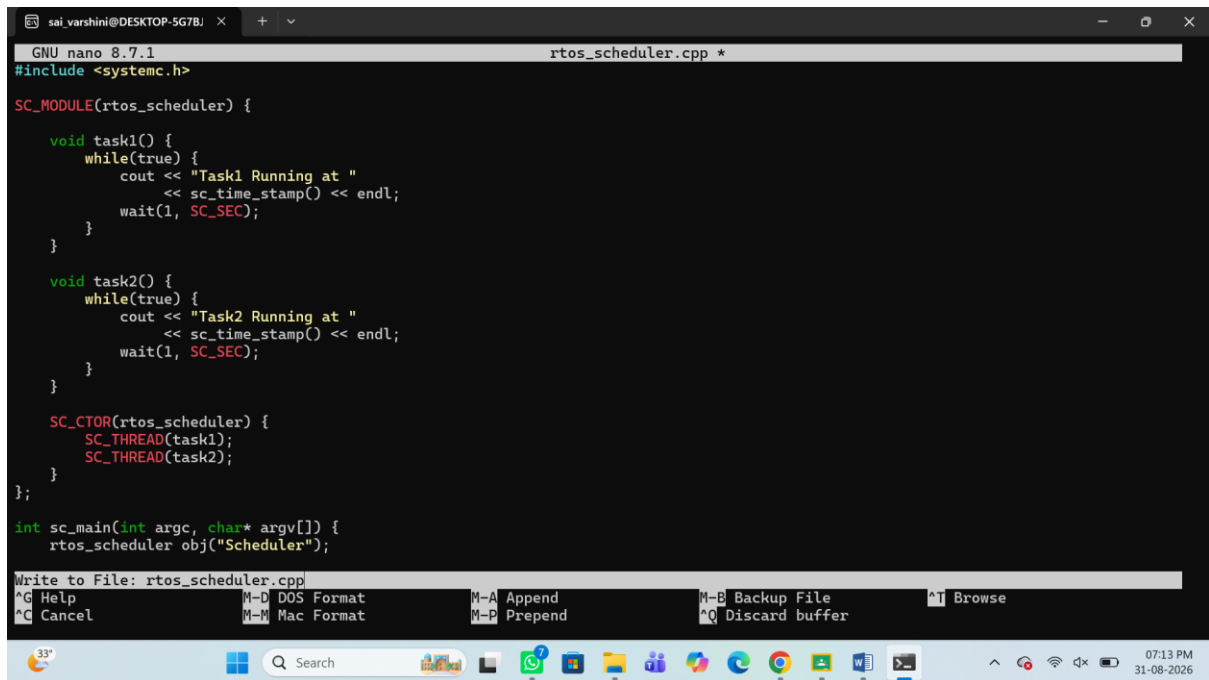


Exp No: 30

RTOS TASK SCHEDULING



```
GNU nano 8.7.1 rtos_scheduler.cpp *
#include <systemc.h>

SC_MODULE(rtos_scheduler) {

    void task1() {
        while(true) {
            cout << "Task1 Running at "
                << sc_time_stamp() << endl;
            wait(1, SC_SEC);
        }
    }

    void task2() {
        while(true) {
            cout << "Task2 Running at "
                << sc_time_stamp() << endl;
            wait(1, SC_SEC);
        }
    }

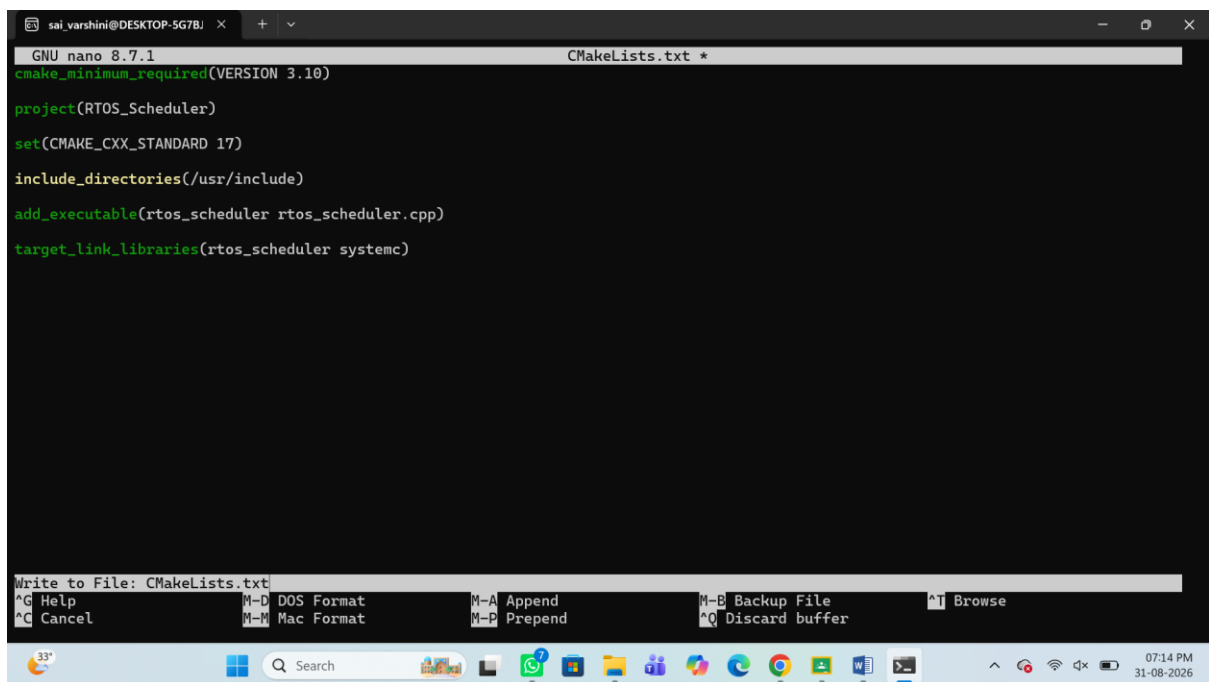
    SC_CTOR(rtos_scheduler) {
        SC_THREAD(task1);
        SC_THREAD(task2);
    }
};

int sc_main(int argc, char* argv[]) {
    rtos_scheduler obj("Scheduler");
}
```

Write to File: rtos_scheduler.cpp

^G Help M-D DOS Format M-A Append M-B Backup File ^T Browse
^C Cancel M-M Mac Format M-P Prepend ^Q Discard buffer

33° Search 07:13 PM 31-08-2026



```
GNU nano 8.7.1 CMakeLists.txt *
cmake_minimum_required(VERSION 3.10)

project(RTOS_Scheduler)

set(CMAKE_CXX_STANDARD 17)

include_directories(/usr/include)

add_executable(rtos_scheduler rtos_scheduler.cpp)

target_link_libraries(rtos_scheduler systemc)
```

Write to File: CMakeLists.txt

^G Help M-D DOS Format M-A Append M-B Backup File ^T Browse
^C Cancel M-M Mac Format M-P Prepend ^Q Discard buffer

33° Search 07:14 PM 31-08-2026

OUTPUT:

```
sai_varshini@DESKTOP-5G7B1 X + v
-- Check for working C compiler: /usr/bin/cc - skipped
-- Detecting C compile features
-- Detecting C compile features - done
-- Detecting CXX compiler ABI info
-- Detecting CXX compiler ABI info - done
-- Check for working CXX compiler: /usr/bin/c++ - skipped
-- Detecting CXX compile features
-- Detecting CXX compile features - done
-- Configuring done (5.3s)
-- Generating done (0.0s)
-- Build files have been written to: /home/sai_varshini/systemc_test/exp29/build/exp30/build
sai_varshini@DESKTOP-5G7B1MJ:~/systemc_test/exp29/build/exp30/build$ make
[ 50%] Building CXX object CMakeFiles/rtos_scheduler.dir/rtos_scheduler.cpp.o
[100%] Linking CXX executable rtos_scheduler
[100%] Built target rtos_scheduler
sai_varshini@DESKTOP-5G7B1MJ:~/systemc_test/exp29/build/exp30/build$ ./rtos_scheduler

SystemC 3.0.2-Accellera --- Nov  5 2025 12:44:46
Copyright (c) 1996-2025 by all Contributors,
ALL RIGHTS RESERVED
Task1 Running at 0 s
Task2 Running at 0 s
Task1 Running at 1 s
Task2 Running at 1 s
Task1 Running at 2 s
Task2 Running at 2 s
Task1 Running at 3 s
Task2 Running at 3 s
Task1 Running at 4 s
Task2 Running at 4 s
Task1 Running at 5 s
Task2 Running at 5 s
sai_varshini@DESKTOP-5G7B1MJ:~/systemc_test/exp29/build/exp30/build$ |
```

